B.Tech 3rd Yr CS II Mid Term

[SET] C

Printed Pages: 12 University Roll Number:

B. Tech II Year, III Semester, Mid Term Exam-2014

AHE-201: Ethics & Values

Time: 01 ½ Hours Max. Marks: 20

Note: This paper consists of two passages. Carefully read the passages and answer the questions having multiple choices. Carefully fill all the information in OMR sheet neatly in the blocks. No partial marking.
Full marks (assigned) will be given for a complete answer that means all correct choices.

Section A

Passage-1 (Attempt All Questions: 1X10= 10 Marks)

Can we exonerate someone from the guilty if someone has acted out of necessity? Queen Vs Dudley and Stephens is one of the most debated cases in the law schools, which had established an example, in the British common law, that necessity is not a defense for a charge of murder. The case is also popular as The Life Boat Case (1884). The English yacht *Mignonette* was to be transported to Australia with crew of four: Tom Dudley, the captain; Edwin Stephens; Edmund Brooks; and Richard Parker, the cabin boy. Case is very much clear to you that Dudley and Stephenson went for trail, while Brooke turned out a witness. Now, judge has to decide whether the killing of Richard Parker and cannibalism done by the three on life-boat for saving their lives was justified. Dudley and Stephens argued that they have done wrong but it was the only way out and was an unavoidable necessary action. Arguing from the necessity they reasoned that necessity was to save the three lives so they killed Parker. Judge: why did you kill parker? Dudley: Parker was orphan and was ill. We other members have family.

We thought if Parker would die nobody would have sorrow but any one of us would die then his family would be in a great pain. One had to die in this scenario and with this necessity parker was the right candidate to be killed. And, from the view point of Consequentialism it was a right decision. Judge came with the point that after four days you guys got rescued and you could have survived for four days without killing Parker then in this possibility killing Parker neither proved necessity not the utility. And, if you had no idea when you will get rescue then killing Parker does not make any sense as there is no guarantee that after subsisting on Parker you will not have to kill another. Killing Richard Parker makes a sense of necessity when there is certainty that the rescue will come and for that time you cannot survive without killing Parker. It was the argument from necessity that makes clear that necessity cannot be sought in the view of uncertainty. Stephenson got disproved on the ground of necessity moved to another argument called argument from intention that we did not want to kill Richard Parker. We wanted to save three lives. But, it is further debatable; was it you will-full action? Stephenson replied no then judge said why did you kill then Parker? You could have killed anyone or could have sacrificed your own life for saving the other three. Stephenson was willing the action as he minded that Parker is ill and is an orphan. The arguments presented by the Dudley and Stephenson based on maximum life saving, survival, sacrifice and life defense did not suffice them to excuse from the charges of murder and cannibalism. The jury decided the sentence to death for both Dudley & Stephenson but that later changed by the interference of the crown into imprisonment. However the case offers a thrilling moral debate on the point whether necessity outweigh the morality. But in this case it was disproved. And, it proved that a murder is murder and it cannot be right even looked by the glaring glasses of morality. The necessity disapproved in this case. However, if we argue

from the consent that suppose if Parker was counseled and he gave the permission to them to kill him to save their lives then does it exonerate them from the charges of murder & cannibalism? We can argue that in prima facie it seems right from the end of Parker that he would have rightly sacrificed to save them, but it cannot be said to be right from the end of Dudley & Stephens to kill the parker. This idea of consent seems right but is highly questionable because we need to answer a question: Do we have right to give such consent to take out lives to others? Do we have complete ownership to our life? We are not the originator of our lives. Yes, we do develop and create it in many ways but it does not mean that we can give ourselves or the others the consent to commit suicide or to killing by the others. Arguing from the view point of categorical value of life we can put forward the reasons that life is valuable in itself without any external facts. And, we have a duty to respect it. Life is not a means to pleasure, joy, successes and good times only but has its own moral worth the way it's naturally offered to us. We have ownership and all rights to our life but in a positive sense to flourish it and enhancing the well being. However, we cannot said to have rights or ownership if we direct negative intentions towards life like moving into depression committing suicide or any other negative desire. The thinkers of the view strongly maintain that the value of life is not comparable and cannot be sacrificed for maximum numbers. They advocate deontological view on morality. Morality for them is not the matter of just utility. They argue that life is valuable in itself. It does not increase or decrease by the numbers of people associated with it or not. For them Parker's life was equally important and not less than any ways even if you compare with the three or you add the lives of their family members too. For such theorists morality does not depend on the consequences. They locate the morality in the nature of an action. The moral worth of an action according to them lies in

the very nature of an action. Hence, they would condemn the case of cannibalism as done out of necessity and would say there cannot be any such necessity beyond the morality. For them the case shows implicit immorality as it compares the value of life with external factors. Necessity means only in a moral framework and it means it has to be moral necessity and not the ordinary necessity. Thus, the necessity is a moral notion and lies within the domain of morality. Well, food is necessity for a hungry person and also for saving the life but killing and eating human for the same reason can never be treated as a moral necessity. Therefore, any argument whatsoever that justifies the Killing of Parker as necessity and right would be demeaning the very sense of morality and thereby, the humanity as well.

Questions:

1) What is the argument from necessity?

a) It was a necessity that compelled Killing Parker.

b) Killing Parker was the only way for the survival.

c) There was no other way other than killing Parker.

d) None of these.

2) Which of the following statements are **wrong** about the case?

a) Dudley murdered Stephens.

b) It was the case of consented cannibalism.

c) Brooke murdered Parker in this case.

d) All above.

3) Necessity cannot be proved in the view of uncertainty. What can we infer from this statement?

a) Uncertainty disapproves necessity.

b) Thomas Dudley was not sure about the rescue.

c) Necessity can be proved in the view of certainty.

d) None of these.

4) Does argument from consent justify the killing of Parker?

a) Consent cannot be ground to exonerate someone from criminal charges.

b) Yes, it does, we have right to give such consent.

c) No, such consent cannot justify the killing.

d) Argument from consent doesn't justify the Killing of Parker.

5) Why did argument from intention fail in the case?

a) They have no intention to kill Parker.

b) Dudley & Stephens planned to kill Parker.

c) They kill Parker knowingly that he was ill and an orphan.

d) All above.

6) Which of the following arguments were **Not** offered by Stephenson in his defense?

a) The argument from justified killing.

b) The argument from maximum lives saving.

c) The argument from uncertainty.

d) None of these.

7) We cannot said to have rights or ownership if we direct negative intentions towards life. What does author argue from this statement?

a) We cannot have negative intentions towards the life.

b) We are not originator of our life.

c) We have only positive rights and ownership.

d) We have only negative rights to life.

8) Necessity cannot go beyond the morality. What can be concluded by this statement?

a) Necessity may go beyond morality.

b) It is a moral necessity.

c) Necessity exists within the morality.

d) All above.

9) Which of the following statements are right about categorical view to this case?

a) Life cannot be sacrificed for maximum numbers.

b) Morality is not a matter of utility.

c) Value of life is not comparable.

d) None of these.

10) For some theorists even if we include the consent life issues cannot be decided on it. What can we infer about these theorists?

a) For them Parker's life was equally important as the life of others.

b) They locate morality in the nature of action.

c) They advocate deontological view of morality.

d) All above.

Section B (Attempt All Questions: 1X10= 10 Marks)

Immanuel Kant, the father of human rights, explains the ideas of individual freedom and respect for human beings in so neat and scientific manner. Being contemporary to Sir Isaac Newton, Kant was very much influenced with Newtonian physics, and he was in full agreement with Sir Newton on the point that nature is governed by the deterministic physical laws. However, this determinism was a sort of challenge for Kant to define, that if we are governed by the deterministic laws then, in what sense.we can be said to be free. Kant by respecting the Newtonian Physics initiates his argument in such a beautiful manner and says that there are two kinds of world: 1) Physical, and 2) Moral. Further, he argues that

we people are governed by the physical laws in so far as we operate in nature (we operate under law of gravitation, in a certain atmosphere consisting of density, pressure and other physical conditions). However, Kant argues this could be a limitation on the animal life because there exists the one and only physical world for them. But, for individual persons, there exists what Kant refers to as moral world. Moral world is such a world that we create for ourselves. And, we can create it by the virtue of human rationality. Moral world is a world of rationality in which the human life is governed by reasons. Kant says it is only human beings who have got mind and its faculties of thinking and reasoning. By having recourse to these faculties of mind, an individual person can intend, resist and can do certain acts and can refrain herself/himself from doing something. Kant argues that a person is free, and free by reason. For example, let's compare two persons one is guided by reason and another who does not use rationality. You can easily understand that the one who follows the reason can explain many events of life but the person who does not follow would be depending either on the other or would be always operating under certain fear or external command. This is the reason why Kant argues that it is rationality through which we get the freedom. Thus, Kantian idea of individual freedom, unlike the Libertarians, is grounded in human rationality. But according to Kantianism: Individual choice should not be sacrificed for larger social purposes. Establishing the idea of individual freedom Kant goes ahead to explain the idea of respect for the human beings. Human beings possess rationality and autonomy they must be respected because they have the capacity to legislate morality for themselves. They can create moral rules for their own lives. This is what Kant calls self-legislation. In other worlds, we can say one can realize the sense of freedom in creating good rules for one's betterment of life.

Kant further tries to explain that morality should be universal and uniform, and should not depend either on personal biasness or external pressure. Morality must be objectively rational in its character that we people can easily understand it without involving the personal reasons, and if we really want to include our personal reasons then we must not personalize the reasons but rationalize them to the extent that they become unbiased, universal and uniform. Kant after explaining the ideas of rationality and freedom further moves to explain the idea of moral accountability and says that we people can be held responsible once we act freely and rationally. However, he makes a very interesting point that it is very much possible that a person does something 'X' and 'X' is right and s/he knows very well that 'X' is right but there is possibility that 'X' is not moral. Really interesting moral fact it is, Kant says that a person who does 'X' but not by his or her moral choice. The person may not have chosen that action or s/he was not free to do any other action. It means the person has performed the action either on the other's choice or under coercion, pressure or force. The action may be right but in Kantian definition it is not moral. So, you do something right but if not on your own choice then your action does not have any moral worth. It cannot be said morally a praiseworthy action. Thus, we can praise or blame someone for his or her actions only in that someone has performed an action with freedom and reason. Kant says moral responsibility can be ascribed to someone who is both a free and rational creature. It is in this sense Kant emphasizes that morality should be free from religion. Kant says morality is an expression of moral law in oneself. Morality must freely evolve in oneself. Kantian description of morality is based on the deontology. Deontological morality is a framework in which moral worth of an action, duty, rule, decision or person lies in the very nature of the action, rule, duty, decision or person.

Unlike, Consequentialism, Kantianism focuses on the means, ways, or the procedures we follow to perform an action. This view holds we should only care the actions not the consequences. We should be careful to the means that we choose to fulfill an end and there is no need to look at the ends. Kant says that even if your right means do not produce the right end do not get depressed as your action carry a moral worth within itself. For example, suppose if someone being student honestly study hard by practicing his duty somehow does not get good marks, then this student, according to Kant, need not be worried and disappointed in so far as his action is morally correct and good. Kant put it in this way "Duty for the sake of duty." Kant maintains universality in his morality by explaining the idea of categorical imperative. He argues in all moral rules there is an imperative but it can be of two kinds: 1) Hypothetical, and 2) Categorical. Hypothetical imperative means the imperative or moral command of an action depends on external conditions. For example, if you want to get good marks then study hard. In this example for a student studding hard is a morally correct thing but it has to qualify a condition, and it is to score the good marks. Categorical imperative is supreme principle of morality. However, categorical imperative means the imperative or moral command of an action is unconditional. For example, do your duty. Kant argues the moral worth of a moral action does not depend on the consequences it produces but depends on our intentions behind our actions. For example, a murderer and a doctor both uses a knife but with different intentions one uses to kill a life and other uses to save a life. Here Kant introduces his beautiful idea of good- will and says that no action can be said to have moral worth unless performed by a sense of good will. What is good-will? Kant replies this question so neatly and says that a will is different from desire. Desires are psychological and represent our emotions and passions. And, they lack

determination that is why our desires get changing all the time. However, will is not purely psychological, it is based on reasons and have got determination as well. Kant says, although will is rational and has got strong determination but without being good a normal will may carry wrong intentions. For example: a strong willed person wants to use his determination in killing someone. This is the reason why Kant argues that our will must follow good intentions that's what he calls good-will. He says without good-will morality cannot be performed and all values say knowledge, power and beauty can be misused. For example, a knowledgeable person can misuse the knowledge for the wrong purposes making cheat and frauds. powerful person can misuse the power by beating a week. Without having good-will we cannot perform moral actions.

Questions:

1) A person 'P' performs an action 'X'. And, 'X' is right. And P knows that 'X' is right. But, his action 'X' can be said to be moral only when

a) 'X' was not chosen by person P.

b) P was free to perform 'X'.

c) P performed 'X' with his own intention.

d) All above.

2) What is moral world according to Kant?

a) A world determined by physical laws.

b) A world where people are governed by reasons.

c) A world of moral reasons.

d) A world of rational persons.

3) Which of the following statements are **wrong** about the Kantian notion of Autonomy(Freedom):

a) We are free because we have rationality.

b) We are free by inclinations (desires).

c) We free by physical nature.

d) None of these.

4) We should respect all human beings. What Kant argues by this statement?

a) They are end in themselves.

b) All human beings like to be respected.

c) They are means like objects and other thing.

d) They can rationally self legislate morality.

5) Someone cannot be held morally responsible for an action if :

a) S/he is motivated through desires.

b) S/he is free to perform an action.

c) S/he is irrational creature.

d) S/he is performing the action not by his own will.

6) What of the following statements are **wrong** about the morality according to Kant?

a) It should evolve in oneself through good results.

b) It should be an expression of moral law in oneself.

c) It should be based on religion.

d) It should be enforced by coercion, force or pressure.

7) Good-will is the only thing that can be said to have moral worth in genuine sense. What we can infer from this statement?

a) Good-will carries good intentions.

b) Good-will is profitable.

c) Good-will is performing action with the spirit of duty.

d) All above.

8) Categorical imperative is the supreme principle of morality. Which of the following statements are right about categorical imperative:

a) The moral command is based on Universality.

b) The moral command in categorical imperative is conditional.

c) The moral command in categorical imperative has nothing to do with the consequence, result or end of an action.

d) The moral command in categorical imperative will bring maximum bad.

9) Which of the following statement is/ are **NOT** correct for deontological morality framework:

a) Moral worth of an action lies in the autonomous reason.

b) Moral worth of an action does not lie in the nature of an action.

c) Moral worth of an action lies in the actor.

d) Moral worth of an action lies in the very sense of duty.

10) What are the basic ideas of Kantian theory?

a) It explains the ideas of individual freedom and respect for human beings.

b) It advocate for rights of individuals.

c) It argues that individual choice should not be sacrificed for larger social purposes.

d) It defines individual freedom on the basis of human nature.

Printed Pages: 2 **University roll no.....**

Mid-Term Examination, 2014-15

Mathematics III (AHM- 201)

Time: 90 Min. **Max. Marks : 20**

Section A

Note: Attempt All Questions. **1X 5 = 5 marks**

(I) Find Fourier sine transform of e^{-x}.

(II) Name the appropriate Fourier transform to solve the partial differential equation

$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ if $u_x(0,t)=0$, $u(x,0)=0, x>0$ and u(x, t) is bounded when x>0, t >0.

(III) Find the Z transform of the sequence { -1, 1, -1,1 ,-1,1}.

(IV) Find the Z-transform of unit impulse function

$$\delta(\zeta)=\begin{cases}1 & \zeta=0\\ 0 & \zeta\neq 0\end{cases}$$

(V) Find the inverse Z -transform of $F(z)=\frac{1}{z-3}$ for $|z|<2$.

Section B

Note: Attempt Any Three Questions. **2X 3 = 6 marks**

(I) Find a Fourier sine integral representation for

$F(x)=\begin{cases}1 & \text{for } 0\le x\le \pi\\ 0 & \text{for } x>\pi\end{cases}$, hence evaluate the integral

$\int_0^\infty \frac{1-\cos\lambda\pi}{\lambda}\sin\lambda t\, dt$.

(II) Find the Fourier transform of $F(x)=\begin{cases}\frac{1}{2a} & \text{for}|x|\le a\\ 0 & \text{for}|x|>a\end{cases}$

(III) Prove that the Z – transform of $f(\zeta)=\frac{1}{\zeta+1!},\zeta\ge 0$ is $z(e^{\frac{1}{z}}-1)$.

(IV) Find the inverse Z- transform of $\frac{z}{z^2+7z+10}$.

Section C

Note: Attempt Any Three Questions. **3 X 3 = 9 marks**

(I) Let $x(t)=e^{-t2},t\ge 0$. Use the procedure that follows to show that its Fourier cosine transform is $X_c(\xi)$, given by

$$X_c(\xi)=\sqrt{\pi}\,e^{-\pi 2\,\xi 2}$$

(II) Solve

$\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2}$ if $u(0,t)=0, u(x,0)=e^{-x}, x>0$ and u (x,t) is bounded when $x>0, t>0$

(III) Find F(x), if its Fourier cosine transform is $\begin{cases}\sin\xi & 0<\xi<\pi\\ 0 & \xi\ge\pi\end{cases}$.

(IV) Find the Z transform of $c^{\zeta}\cos(\tau\zeta),\zeta\ge 0$.

II Term Examination, 2014-15
Course: B. Tech. (CS), Year: II, Semester: III

Data Structure using C (CSE 211)

Uni. Roll No:

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Differentiate between complete binary tree and full binary tree.
II. What is the balance factor of AVL tree? Give example.
III. What Adjacency Matrix. Give example.
IV. Give any Two properties of a B.Tree.
V. What are the advantages of a threaded binary tree?

Section B

Note: Attempt any THREE Questions
(2x3=06)

I. Show that for the perfect binary tree of height h containing 2^{h+1}-1nodes.
II. Draw a Binary search tree for the following input list 60,25,75,15,50,66,33,44.Trace the algorithm to delete the nodes 25,75,44 from the tree.
III. Show the result of inserting 2,4,1,5,9,3,6,7.17,8 in to an initially empty AVL Tree.
IV. The Inorder and Preorder traversals of T yields the following sequence of nodes:

Inorder: C D E B G H F K L P Q M N J A
Preorder: A B C D E F G H J K L M P Q N

Draw the tree T

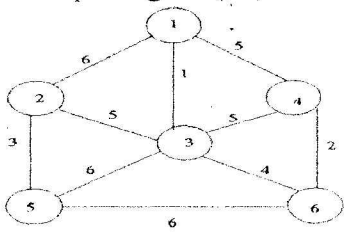
Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Insert following elements from an initially empty BTree of minimum degree 2 .

Values: 45, 23, 29, 85, 92, 7, 11, 35, 49,51,20,5,91,64.Delete 45,5,20,29.

II. Apply Prim's algorithm on the following graph to find minimum cost spanning tree.(Show all the steps)



III. Create a Huffman tree using the following information:

Data item:	A	B	C	D	E	F	G	H
Weight:	22	5	11	19	2	11	25	5

Construct the tree showing each step separately.

IV. Differentiate between DFS and BFS graph traversal with the help of an example graph.

Printed Pages-2 **University Roll No.....**

B. Tech. II Year III Semester, Second Term Examination, 2014-15

CSE-212: Computer Organization

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Symbols have their usual meanings.

Section-A

Note: Attempt all Questions **(1*5=5)**

I. Let SP=0000000 in the stack. How many items are there in the stack if Full=1 and Empty=0?

II. Write Reverse polish notation of A+B*(C*(D-E)).

III. Define Micro-routine and Micro-Instruction?

IV. Consider n+ k bits one address instructions with k-bits operation code and n-bits address. How many maximum address it can specify and how many maximum operation it can perform?

V. Explain Mask logic operation with example.

Section-B

Note: Attempt any three questions **(2*3=6)**

I. A computer uses a memory unit with 128K words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four fields: an operation code field, a mode field to specify one of the five addressing modes, a register address field to specify one of the 100 processor registers, and a memory address. Specify the instruction format and the number of bits in each field if the instruction is in one memory word. How many maximum operations system can perform?

II. An instruction is stored at location 151 with its address field at location 152. The address field has the value 300. A processor register R1 contains the number 200. Evaluates the effective address if the addressing mode of the instruction is

a) Direct b) Immediate c)Relative d)Index with R1 as the index register

III. Differentiate between various implementation methods of Control Unit.

IV. Starting from initial value of R=11000111, Determine the sequence of binary values in R after a logical shift left followed by arithmetic shift left followed by circular shift right followed by logical shift right.

Section-C

Note: Attempt any three questions **(3*3=9)**

I. Evaluate given expression by using general register organization with two address instruction format and stack organization.
X = (A + B - C *(D – E * F)) / (G * H + K)

II. Write the sequence of control steps with control signal required for the execution for each of the following instruction:

a) Add the immediate number A to Register R1 and Store result in R1.

b)Add the contents of Memory location B to R1 and Store result in R1.

III. Explain various Micro-instruction Addressing techniques.

IV. Write Short notes on following.

a) Types of Micro-Instruction

b) Micro-Instruction with Next Address Field

Mid Term Examination, 2014-15

Discrete Mathematics (CSE 213)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. A box contains two white socks and two black socks. Two socks are drawn at random. Find the probability that they are of same color.

II. The probability that A hits the target is 1/6 and the probability that B hits a target is 2/3. They both fire at a target. Find the probability that neither hits the target.

III. Write the negation of the following statement using propositional calculus: **"If the triangle is equilateral, then it is equiangular."**

IV. Write the contrapositive of the following statement using propositional calculus: **"The home team wins if it is raining."**

V. Show the following equivalence without using truth table:

$$(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$$

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Prove that $\sqrt{3}$ is an irrational number by the method of contradiction.

II. A, B, C are three mutually exclusive and exhaustive events associated with a random experiment. Find P(A), it being given that P(B) = 3/2 P(A) and P(C) = ½ P(B)

III. A box containing 5000 IC chips, of which 1000 are manufactured by company A and the rest by company B. 10% of the chips made by company A and 5% of the chips by company B are defective. If a randomly chosen chip is found to be defective, find the probability that it came from company A.

IV. Determine whether $[\{(p \vee q) \rightarrow r\} \wedge (\neg p)] \rightarrow (q \rightarrow r)$ is a tautology, contingency or contradiction.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Use mathematical induction to show that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer value of n.

II. Obtain the PDNF and PCNF of $(p \wedge \neg q \wedge \neg r) \vee (q \wedge r)$.

III. Show that premises $P \rightarrow Q$, $R \rightarrow S$, $\neg Q \vee \neg S$, $\neg \neg P$ and $(T \wedge U) \rightarrow R$ imply the conclusion $\neg (T \wedge U)$ using rules of inference theory.

IV. Rewrite the following statements using quantifiers, variables and predicate symbols by specifying universe of discourse:

d) All men are genius.

e) Not all students are intelligent.

f) No well-behaved person is quarrelsome.

Second Term Examination, 2014-15
Course: B.Tech. (CSE), Year: II, Semester: III
Object Oriented Programming (ITE 211)

Uni. Roll No:

Time: 1.5 Hrs. **Total Marks: 20**

Instructions:-

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

1. What will be the output of the following program?

```
class Vehicle{
  Vehicle (){System.out.println("Vehicle is created");}
}
class Bike6 extends Vehicle{
  int speed;
  Bike6 (int speed) {
    this.speed=speed;
    System.out.println(speed);
  }
  public static void main(String args[]){
    Bike6 b=new Bike6(10);
  }
}
```

II. What will be the output of the following program?

```
class A{
 int a=40;
 public static void main(String args[]){
  System.out.println(a);
 }
}
```

III. What will be the output of the following program?

```
class Print{
 public static void main(String args[])
 {
  String s="GLA";
  s.concat(" University");
  System.out.println(s);
 }
}
```

IV. Why use Java interface?

V. Do you need to include **Java.lang** explicitly? Show how to include all of the classes in a package called **toolpack.**

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Create an abstract base class shape with two members base and height, a member function for initialization and a pure virtual function to compute area ( ). Derive two specific classes Triangle and Rectangle which override the function area ( ). Use these classes in a main function and display the area of a triangle and a rectangle.

II. Compare and contrast.

(a) Method Overloading and Method Overriding.

(b) Abstract class and Interface

III. Explain the Life cycle of a Thread? Write a java program to implement Runnable interface to create a thread.

IV. When do we declare a variable, method, or class final? Explain with program.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. How many catch blocks and finally blocks can we used with try block? Write a program for user defined exception that checks the internal and external marks if the internal marks is greater than 20 it raise the exception" Internal marks is exceed" , if the external marks is greater than 80 exception is raised and display the message " The external marks is exceed". Create the above exception and use it in your program.

II. Explain the types of Layout Managers in GUI programming

III. Define package. What is the usage of javac –d option? Write the programs for creating package and importing package.

IV. What is an Applet? Write the differences between an Applet and Application? Explain the life cycle of an Applet.